KUM DONALSIEN AKWO

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PROFESSIONAL PROFILE

B.Tech Computer Engineering graduate specialising in Networks and Security at the College of Technology, University of Buea, with a strong passion for cybersecurity and cyberspace as a whole. Skilled in cybersecurity, network security engineering, AI/ML-based and hybrid intrusion detection systems and other network monitoring and auditing tools. Holds 2 years of hands-on technical experience in network installation, configuration, troubleshooting and security with Telnet Cameroon. Designed and built Bastion IDS, a full hybrid intrusion detection system combining signature-based detection, machine learning ensemble models, deep learning, and unsupervised learning models for real-time threat detection. Proven leader as Program Delegate for the Computer Engineering Department, under the Computer Networks and Systems Maintenance option.

EDUCATION

COLLEGE OF TECHNOLOGY, UNIVERSITY OF BUEA Oct 2023 - Aug 2026

Bachelor of Technology (B.Tech) Computer Engineering (**First Class Honours**) **(Coursework Completed)**

3.73/4.0

GCE ADVANCED LEVEL June 2023 Session

General Certificate of Education (GCE) Board, Cameroon

- Biology, Chemistry, Pure Mathematics with Mechanics, Physics, Computer Science

SKILLS

- Cyber security (Kali Linux, Wireshark, Nmap, Snort, Suricata, IDS/IPS, Ethical Hacking, Penetration Testing, Fortigate, Data Security, Security Operations, Cyber Threat Intelligence, Social Engineering, Information Security)
- Networking (Cisco IOS, Cisco Switches & Routers, OSPF, BGP, IS-IS, RIP, Network Design, Wireless Networking, Network Architecture, IP Addressing)
- Cloud & AWS (AWS EC2, S3, VPC, IAM, CloudFront, CloudWatch, Route 53, VPC design, cloud networking, cloud security)
- Programming (Python, JavaScript, C, MySQL)
- AI & Machine Learning (Scikit-learn, Keras/TensorFlow, Random Forest, XGBoost, CatBoost, Isolation Forest, Autoencoders, Deep Neural Networks, pandas, LLM Evaluation and Benchmarking)
- Frameworks & Tools (FastAPI, Flask, React, Tailwind CSS, Electron, Vite, Docker, Git, VirtualBox, MITRE ATT&CK)
- Operating Systems (Windows, Kali Linux)
- Productivity (Microsoft Office Suite)

EXPERIENCE

TELNET CM August 2024 - September 2024

Intern

- Completed a structured 3-month internship focused on network installation, configuration and security operations
- Gained hands-on experience with physical Cisco switches, routers, Fortigate firewalls and enterprise network infrastructure
- Configured routing protocols (OSPF, BGP, IS-IS, RIP), wireless access points and secure remote access solutions
- Deployed IP-based security surveillance systems and Starlink satellite internet infrastructure on client sites
- Recognised and awarded Best Intern for the batch by TELNET CAMEROON management

TELNET CM October 2024 - Present

Support Staff

- Provides ongoing technical support and network engineering assistance across client projects
- Assists in planning and executing corporate network infrastructure deployments for enterprise and institutional clients
- Applies cybersecurity monitoring best practices and contributes to network security posture assessments
- Supports installation and configuration of Cisco equipment, Fortigate firewalls and IP surveillance systems

TELNET CM July 2025 - September 2025

Senior Intern & Intern Instructor

- Returned for a second formal internship in a senior capacity with expanded technical and leadership responsibilities
- Served as lead instructor and mentor for a new batch of junior interns, delivering training in network configuration, troubleshooting and cybersecurity best practices
- Supervised junior intern lab sessions covering Cisco IOS configuration, routing protocols and network security concepts
- Continued developing advanced skills in corporate network deployments and field security implementations

PROJECTS

Bastion IDS, A Hybrid Intrusion Detection System (2025-2026)

Designed and developed a multi-layer hybrid IDS as a capstone project.

Bastion IDS features a four-layer detection mechanism: Layer 1 is the signature engine (4ET-Open ruleset, 47,357 signatures), Layer 2 is a machine learning ensemble (Random Forest, XGBoost, CatBoost), Layer 3 is a deep neural network specialist (residual MLP) and Layer 4 is an anomaly detection engine. Developed as a standalone desktop application for real-time network monitoring.

Technologies: *Python, FastAPI, React, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, Wireshark, Snort*

Stop Arnaque 237, Open Cameroon Scam Dataset (2026)

Created and released an open, labelled dataset of real scams in Cameroon, crowdsourced from the people who receive them and published with a Zenodo DOI under CC BY 4.0. Real scam messages sit beside the genuine mobile-money messages they imitate, in English, French and Pidgin, anonymized and labelled for scam-detection models.

Technologies: *Python, pandas, NLP, Data Anonymization, Hugging Face, Zenodo*

Does the Answer Survive Disagreement, LLM Sycophancy Benchmark (2026)

Designed and published an open benchmark measuring whether a model keeps a correct answer when a user asserts a wrong one, with a control arm that separates caving from sensibly updating on evidence. Ran it across 12 models from six labs on Kaggle Benchmarks and published a technical write-up on DEV.

Technologies: *Python, pandas, Data Visualization, Kaggle Benchmarks*

FairGlass, Proof of Fair AI Hiring (2026)

Built on the Midnight blockchain during an MLH hackathon. A Compact smart contract proves that an AI hiring decision followed an agreed fairness policy, issuing a verifiable receipt when the rules hold and refusing one when a forbidden attribute changes the outcome, all without exposing candidate data. Contributed the backend, scoring logic and the contract bridge.

Technologies: *Python, Flask, Midnight (Compact smart contracts), Cryptographic Commitments*

Evaluation and Recommendation of Network Monitoring Tools (Aug - Sep 2024)

Evaluated industry-standard IDS/IPS and network monitoring tools to identify and prevent cyber threats. Produced formal analysis and recommendations report to enhance organisational security posture.

Technologies: *Wireshark, Snort, Suricata, Kali Linux, Nmap, Security Onion, Kitsune*

Chat Application and File Transfer System (Dec 2024 - Jan 2025)

Built a real-time multi-client chat application with secure file transfer capabilities using Python socket programming.

Technologies: *Python, Socket Programming*

PUBLICATIONS

Tatsachou Ngningaye D., **Kum Donalsien A.**, Gamom Ngounou Ewo R. C., and Boum Alexandre T. "Bastion IDS: Layer Contribution, Threshold Sensitivity, and Consensus Rule Design in a Deployed Hybrid Intrusion Detection System." Journal of Cyber Security Technology (Taylor & Francis Group), 2026. Submitted.

Kum, D. A. (2026). Stop Arnaque 237: Cameroon Scam Reports Dataset (Version 1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.23287618

LANGUAGE

- English (Fluent), French (Basic)

ACTIVITIES

- 237HackFest Capture the Flag (CTF) Competition, organised by Hack The Box Cameroon, gained hands-on experience solving real-world cybersecurity challenges
- Major League Hacking in partnership with the Midnight network, 48-hour hackathon building data protective, privacy-first applications. Developing dApps and tools with ZK technology that put users back in control of their personal data.
- Program Delegate, Computer Engineering Department, Option: Computer Networks and Systems Maintenance, University of Buea, key liaison between students and university administration

SOFT SKILLS

- Leadership
- Teamwork
- Critical Thinking
- Problem Solving
- Effective Communication
- Adaptability
- Time Management
- Attention to Detail
- Resilience

CERTIFICATIONS

- Certified in Cybersecurity (CC), ISC2 (September 2026, valid to September 2029)
- Fortinet Certified Fundamentals (FCF) in Cybersecurity, Fortinet (July 2026, valid to July 2028)
- Cisco Introduction to Cybersecurity, Cisco Networking Academy (September 2025)
- Cisco Networking Academy Capture the Flag 2026, Cisco (May 2026)
- Cyber Crime, Humanitarian Leadership Academy (July 2024)
- Cyber Security, Humanitarian Leadership Academy (July 2024)

ACHIEVEMENTS & AWARDS

- Academic Excellence Recognition, consistently maintaining an excellent CGPA throughout the degree programme
- Leadership skills award from the College of Technology for demonstrating outstanding leadership capabilities in school
- Best Intern Award, Recognised by TELNET CAMEROON management as best intern for the August-September 2024 batch